



Entropy-regulated cross-modal generative fusion for multimodal network intrusion detection

Xiangbin Wang^{a,b}, Qingjun Yuan^{a,b}, Wentao Yu^{a,c}, Qianwei Meng^{a,b}, Siqi Lu^{a,b}, Wenqi He^{a,b}, Chunxiang Gu^{a,b}, Yongjuan Wang^{a,b,**}

^a Information Engineering University, Zhengzhou, 450001, Henan, China

^b Henan Key Laboratory of Network Cryptography Technology, Zhengzhou, 450001, Henan, China

^c Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China

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ABSTRACT

With the increasing popularity of network encryption protocols, analyzing encrypted traffic has become a significant challenge for network security. In this context, deep learning methods have been widely applied to intrusion detection and traffic classification tasks due to their powerful feature extraction capabilities. However, these methods still face two main limitations: relying on unimodal feature extraction, which ignores the multimodal characteristics of network traffic, or adopting simple static fusion strategies, which may fail to capture the complex semantic associations between different modalities. These limitations make it challenging for models to detect sophisticated attacks concealed within otherwise normal encrypted traffic. To address this challenge, this paper introduces an Entropy-Regulated Cross-Modal Generative Framework for Intrusion Detection (ER-CMGI), which combines the generative power of diffusion models with dynamic information-theoretic optimization techniques. The framework integrates variational autoencoders (VAEs) with a lightweight diffusion model for multimodal feature extraction and generation. It implements an adaptive fusion mechanism using a hybrid entropy-based approach that combines both traditional low-entropy priority and inverse entropy weighting through learnable mixing coefficients. The cross-modal generation consistency is achieved through a lightweight diffusion model, which enables self-supervised learning via direct cross-modal generation and comparison. This design enhances semantic alignment across heterogeneous modalities through cross-modal generative learning. Experimental results show that the proposed model achieves F1 scores of 99.12% and 97.81% on two datasets, respectively. This study presents a dynamically adaptive and entropy-guided framework for intrusion detection in network environments, which shows effectiveness in capturing complex attack patterns. By integrating dynamic feature fusion with cross-modal semantic modeling, the framework enhances detection accuracy and interpretability, offering a promising approach for improving network security under evolving threat scenarios.

1. Introduction

With the widespread adoption of network encryption protocols (e.g., TLS 1.3, QUIC), traditional intrusion detection systems (IDS) based on plaintext analysis are facing unprecedented challenges in effectiveness [1–3]. These conventional methods – including port scanning, fixed signature matching, and superficial traffic analysis – are increasingly ineffective. Attackers now exploit encrypted channels to conceal malicious payloads, which renders these approaches insufficient for modern threats.

This challenge has driven the development of deep learning-based detection methods. Researchers expect deep neural networks to automatically extract implicit features from encrypted traffic [4–6]. Early studies applied Convolutional Neural Networks (CNNs) to analyze traffic statistical features [7,8]. Others used Recurrent Neural Networks (RNNs) to model temporal patterns [9,10]. Additionally, graph neural networks (GNNs) have been leveraged to capture non-Euclidean spatial features in network traffic [11,12].

However, unimodal deep learning approaches have fundamental limitations. First, their reliance on single-source feature representations hinders comprehensive understanding of complex attacks. Second, they

* Corresponding author at: Information Engineering University, Zhengzhou, 450001, Henan, China.

** Corresponding author at: Information Engineering University, Zhengzhou, 450001, Henan, China.

E-mail addresses: Moskyes@outlook.com (X. Wang), gcxyuan@outlook.com (Q. Yuan), pinkywyj@163.com (Y. Wang).

often neglect the multidimensional characteristics inherent in cyber attacks. Third, these methods fail to recognize cross-modal correlation patterns in covert attacks. For example, DDoS attacks exhibit burst connection patterns at the flow level, whereas encrypted malware may reveal protocol anomalies at the byte level. Such heterogeneous, cross-layer features cannot be adequately captured by unimodal approaches. This limitation highlights opportunities for applying multimodal learning in intrusion detection systems [13–16].

Existing multimodal intrusion detection approaches, despite their preliminary progress, reveal two structural deficiencies. First, current frameworks employ static weight fusion strategies. These strategies predetermine fixed contribution ratios across modalities. This rigid design overlooks the dynamic nature of traffic feature uncertainty. Specifically, the information entropy of different modalities in encrypted traffic inherently fluctuates with protocol types and encryption intensity. Second, contemporary methods attempt to integrate multimodal representations. However, they inadequately address the inconsistency in the latent space between heterogeneous features. For example, discrete byte sequences and continuous statistical metrics differ fundamentally in their representation. This misalignment results in suboptimal cross-modal alignment. Such shortcomings critically undermine detection accuracy under adversarial conditions.

The advancement of generative artificial intelligence (GenAI) models offers new insights to address these bottlenecks. By constructing cross-modal generative channels, the semantic consistency in latent space can enhance the robustness of feature fusion. For instance, diffusion models have proven effective in modeling complex mapping relationships between high-dimensional discrete data (e.g., network byte sequences) and low-dimensional continuous features (e.g., traffic statistics) [17,18]. The progressive denoising process inherent to diffusion models aligns intrinsically with the incremental feature extraction of anomalous signals in network traffic. Their Markov chain structure can model hierarchical correlations between raw bytes and packet length sequences, providing new possibilities to overcome the mode collapse issue in traditional generative models. However, their potential in cybersecurity remains underexploited.

The aforementioned challenges indicate that constructing an effective encrypted traffic intrusion detection system requires simultaneously meeting two core requirements: (1) the capability to adapt to multimodal feature uncertainties across different encryption scenarios dynamically and (2) the establishment of consistent mapping relationships in heterogeneous feature spaces to capture cross-modal attack patterns. While traditional generative models possess the potential for cross-modal transformation, their fixed generative paradigms struggle to adapt to dynamically changing network environments and lack mechanisms for quantifying information uncertainty.

To address these issues, this paper proposes an Entropy-Regulated Cross-Modal Generative Framework for Intrusion Detection (ER-CMGI). This framework systematically addresses two fundamental limitations in existing methods by integrating the probabilistic generative capabilities of diffusion models with dynamic information-theoretic fusion strategies. First, it achieves adaptive adjustment of multimodal contributions through an entropy-guided fusion mechanism that dynamically weights different modalities based on their information content. Second, it establishes semantic alignment between heterogeneous modalities using cross-modal generation consistency constraints implemented through diffusion models. These complementary innovations enable the model to simultaneously capture both deterministic patterns and high-entropy anomalies in encrypted traffic, significantly enhancing the detection capability for sophisticated attacks. Our main contributions are as follows:

1. We have developed an end-to-end multimodal feature learning system for network traffic specifically designed to handle the complexity of encrypted traffic. This system combines a variational autoencoder (VAE) with a lightweight diffusion model, enabling it to simultaneously process various modal features such as byte sequences and packet

length sequences of traffic. This architecture, while extracting the hidden patterns in network traffic, effectively overcomes the limitations of traditional methods in handling heterogeneous traffic features. In particular, the fusion strategy we designed, based on information entropy theory, establishes a unified processing framework for different types of network features, thereby enhancing the detection ability of encrypted malicious traffic.

2. We have proposed an adaptive feature fusion mechanism based on differential entropy, specifically aimed at solving the problem of optimal weight allocation for network traffic features. Unlike traditional methods that simply favor low-entropy features, our hybrid entropy strategy can simultaneously utilize both deterministic network behavior features and high-entropy features that may contain anomalous traffic patterns. This mechanism dynamically adjusts fusion weights based on the information content of each modality, effectively addressing the heterogeneity and diversity of network traffic while providing an interpretable basis for security decision-making.

3. We have designed a cross-modal generation consistency framework, utilizing a lightweight diffusion model as a semantic alignment tool between different network traffic modalities. Rather than using generative models for data augmentation, we employ them to enforce cross-modal semantic consistency. By comparing the consistency between original modality data and cross-modal representations generated by the diffusion model, we guide the encoder to learn more discriminative feature representations. This approach establishes semantic mappings between network traffic features such as byte sequences and packet length distributions, improving the encoder's ability to capture traffic patterns through self-supervised cross-modal generation.

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 details the ER-CMGI architecture, Section 4 presents comprehensive experimental evaluations, and Section 6 concludes the research.

2. Related work

2.1. Unimodal deep learning for network traffic recognition

Deep learning has emerged as a dominant paradigm in network traffic classification, particularly for analyzing encrypted traffic, where traditional payload inspection is no longer viable. These methods typically operate on a unimodal input – such as raw packet bytes, timing sequences, or statistical features – and leverage models like CNNs, RNNs, GNNs, and Transformers to classify traffic types (e.g., HTTP, FTP, VoIP) or detect anomalies [19–24]. In recent years, distributed approaches have also been explored to improve scalability and efficiency, especially in large-scale or real-time traffic analysis scenarios [25–27]. Compared to rule-based approaches, deep learning models offer greater flexibility through automatic feature extraction, reduced reliance on manual engineering, and firm performance on large-scale datasets.

Existing approaches can be broadly categorized into three directions: (i) feature-based methods that rely on manually extracted statistics (e.g., flow duration, packet size distributions), (ii) end-to-end models that learn hierarchical representations directly from raw traffic data (e.g., byte sequences, header fields), and (iii) distributed frameworks that integrate deep learning with federated learning to enable collaborative training while preserving privacy [25,28,29]. While these methods have demonstrated effectiveness in specific settings, there remains room for improvement in terms of generalization across diverse scenarios, particularly when dealing with multimodal information integration.

Feature-based models, while computationally efficient, may face challenges in encrypted scenarios where discriminative signals in statistical metadata are limited. End-to-end models, although capable of capturing fine-grained patterns in raw data, typically focus on single modalities and may not fully exploit the complementary information available across different data representations. For example, raw

bytes excel at capturing semantic content but may have limitations in representing temporal dynamics, while packet length sequences effectively model behavioral patterns but provide less detailed payload information.

Building upon these insights, our proposed multimodal framework integrates two complementary modalities: raw packet bytes and packet length sequences. Through an entropy-guided adaptive fusion mechanism and cross-modal generation consistency loss, our approach systematically combines the strengths of both modalities to enhance overall performance. This integration is supported by a theoretically grounded framework that employs Kullback–Leibler (KL) divergence to ensure alignment between latent representations, enabling dynamic adaptation to varying network conditions. The resulting method offers improved classification accuracy, enhanced robustness, and better interpretability, making it well-suited for real-world security-sensitive applications. Rather than replacing existing approaches, our framework provides a complementary perspective that demonstrates the value of principled multimodal integration in network traffic analysis.

2.2. Multimodal learning for network traffic recognition

Recent advances in multimodal deep learning have been applied to encrypted traffic classification, aiming to improve accuracy by integrating complementary information from multiple data sources. Early approaches such as AppNet [30] and MIMETIC [14] combine raw packet bytes and packet length sequences using CNNs and recurrent networks (e.g., LSTM, GRU) to extract modality-specific features before applying simple fusion techniques like concatenation. These methods were later extended to multi-task frameworks such as DISTILLER [15], which performs joint classification across multiple traffic attributes. More recently, Lin et al. proposed PEAN [16], leveraging a Transformer encoder and bidirectional LSTM to jointly model byte sequences and packet length dynamics.

A concurrent line of work introduces robustness considerations into multimodal fusion. For instance, the Anti-Noise Network (AN-Net) [31] proposes a representation enhancement strategy that operates at the feature level rather than the input level, due to the numeric nature of encrypted traffic. This method uses a probability masking scheme to randomly discard or scale modal representations during the training process, forcing the model to learn more robust patterns. In addition, AN-Net uses mutual information estimation to select information modalities before fusion, thereby filtering out irrelevant or noisy modalities. The final fusion is performed through average pooling over the enhanced representations.

Despite their effectiveness, current multimodal approaches, including AN-Net, exhibit several key limitations. First, most methods employ static fusion strategies – such as feature concatenation or fixed-weight combinations – that cannot adapt to varying input characteristics. Second, these models process modalities independently during feature extraction, missing opportunities to exploit cross-modal correlations and complementary semantic relationships inherent in network traffic data. Third, existing approaches generally lack principled mechanisms for uncertainty quantification, limiting their ability to provide confidence estimates for predictions, particularly in ambiguous or adversarial scenarios. Finally, while AN-Net incorporates practical noise-robust techniques, its fusion mechanism relies on simple average pooling without dynamic adaptation based on input context or information-theoretic principles.

Our proposed framework addresses these limitations through several key innovations. We introduce an entropy-guided adaptive fusion mechanism that dynamically integrates raw bytes and packet length sequences based on their information content and contextual reliability. The framework incorporates a cross-modal generation consistency loss that explicitly models the latent interactions between modalities, while KL divergence-based alignment ensures theoretical rigor in representation learning. This design provides several advantages over existing

Table 1

Comparison of multimodal network traffic analysis methods.

Method	Core ideas	Strengths	Fusion strategy
AppNet [30]	Hybrid CNN-RNN for TLS traffic analysis	End-to-end learning; Spatial–temporal patterns	Feature concatenation
MIMETIC [14]	Multimodal DL for encrypted mobile traffic with data heterogeneity	Overcomes single-modality limits; Mobile scenarios support	Intra- and inter-modality learning
DISTILLER [15]	Multitask multimodal framework with joint optimization	Multi-objective optimization; Unified framework	Late fusion with fixed weights
AN-Net [31]	Anti-noise architecture with MI-based modality selection	Noise robustness; Intelligent filtering	Average pooling after selection
ER-CMGI	Entropy-guided cross-modal generation with diffusion models	Dynamic adaptation; Theoretical rigor; Semantic alignment	Adaptive entropy weighting

methods: enhanced robustness to partial modality corruption, improved generalization through principled uncertainty quantification, and reduced computational overhead via adaptive processing. By grounding the fusion strategy in information theory, our approach offers both practical improvements and theoretical guarantees for multimodal network traffic analysis.

Table 1 provides a comprehensive comparison of our ER-CMGI framework with representative multimodal network traffic analysis methods, highlighting the essential differences in core ideas, strengths, and fusion strategies.

2.3. Generative methods in network traffic modeling

Generative modeling has also been applied in network traffic analysis, including basic data augmentation, fine-grained tracking reconstruction, and other aspects. These generation methods provide practical solutions to fundamental challenges such as data scarcity, privacy constraints, and class imbalance by synthesizing real traffic samples that closely resemble real network behavior.

One line of research explores the use of generative models for data augmentation in intrusion detection systems. For example, Andresini et al. [32] proposed a GAN-based framework that transforms traffic flows into image-like representations to generate synthetic attack samples. This approach effectively mitigates the imbalance between benign and malicious traffic, thereby improving generalization and detection accuracy for previously unseen attacks.

Recent advances have further expanded the application of GANs in intrusion detection systems, addressing specific challenges across various network environments. GAN-LSTM architectures have been developed for fog computing environments to address computational constraints while maintaining detection accuracy [33]. In mobile ad hoc networks (MANETs) [34], Multi-View Consistent GANs (MVC-GANs) approaches generate consistent multi-view representations for classifying various attack types. To address data imbalance, multicritics GAN clustering-based frameworks employ multiple discriminators to improve synthetic data quality [35]. Furthermore, adaptive network intrusion detection systems utilize stacked Conditional Tabular GANs (CTGANs) to address data drift issues in real-world deployments, preventing catastrophic forgetting during model updates [36].

More recent efforts have focused on generating high-fidelity traffic traces using diffusion models. Sivaroopan et al. introduced NetDiffusion [37], an end-to-end system that applies diffusion-based generation

to network traffic synthesis. Compared to GANs, this method demonstrates better sample diversity and fidelity, making it particularly useful when access to real attack data is limited due to privacy or legal concerns.

Another emerging direction aims at reconstructing detailed packet-level traffic from coarse-grained counter data — a task referred to as network traffic super-resolution (TSR). Wang et al. proposed ZOOMSYNTH [38], which introduces a hierarchical architecture called CLTM, composed of multiple Granular Traffic Transformers (GTTs) operating at different temporal resolutions. When available, rule-based constraints such as ACLs are incorporated to guide semantically meaningful generation. Experimental results show that ZOOMSYNTH can produce high-quality traffic traces and improve performance on downstream tasks such as anomaly detection and service recognition.

While existing methods primarily focus on generating realistic traffic data for augmentation, addressing data imbalance, or handling data drift challenges, our work explores a fundamentally different direction: learning better multimodal representations through cross-modal generation consistency. Unlike previous approaches that use generative models primarily for data synthesis or preprocessing, we integrate lightweight diffusion models directly into the representation learning process to enforce semantic alignment between heterogeneous modalities. We introduce a cross-modal generation consistency loss that encourages one modality's encoder to generate features compatible with another modality's representation space, thereby promoting semantic coherence across modalities. This approach does not aim to replace existing generative frameworks for data augmentation but rather complements them by emphasizing the role of internal cross-modal representation learning in multimodal settings. This design ensures that each modality's encoder captures not only its intrinsic characteristics but also maintains semantic compatibility with other modalities, resulting in more robust and transferable feature representations for intrusion detection tasks.

3. Methodology

This section presents ER-CMGI, a multimodal framework for intrusion detection that effectively learns and integrates heterogeneous network traffic features. As illustrated in Fig. 1, the model consists of five core components: traffic preprocessing, latent representation learning, cross-modal consistency, entropy-guided fusion, and classification. The processing pipeline begins by segmenting raw network traffic into structured modalities, which are then encoded into probabilistic latent representations using a VAE-based architecture. A lightweight diffusion model facilitates cross-modal consistency learning, where latent representations from different modalities are compared and aligned to ensure semantic coherence across modalities. An adaptive fusion mechanism subsequently integrates these aligned representations by leveraging entropy estimates to dynamically weight each modality based on its information content. Finally, a classification head performs intrusion detection using the fused representations, achieving high detection accuracy and robust generalization. Each component is described in detail in the following subsections. The key mathematical notations used throughout this paper are shown in Table 2.

3.1. Network traffic preprocessing

To enable effective identification of network traffic, it is essential to preprocess the raw PCAP data. Preprocessing not only completes the traffic segmentation but also removes background traffic that is not significant for the classification task, and processes the traffic into a format suitable for the input of deep learning models. Importantly, the encryption status of the traffic does not impact the effectiveness of the preprocessing pipeline. Our preprocessing framework consists of four key components: Background Traffic Removal, Flow Segmentation, Irrelevant Field Processing, and Multimodal Feature Extraction.

Table 2

Key mathematical notation and symbols.

Symbol	Definition
$\mathcal{X} = \{x_b, x_l\}$	Multimodal network traffic sample
$x_b \in \mathbb{R}^{S \times F}$	Raw byte sequence modality
$x_l \in \mathbb{R}^S$	Packet length sequence modality
$z_b, z_l \in \mathbb{R}^d$	Latent representations for byte and length modalities
$\mu_b, \mu_l, \sigma_b^2, \sigma_l^2$	Mean and variance vectors of latent Gaussian distributions
$H(z)$	Differential entropy of latent representation z
$\bar{H}(z)$	Sample-averaged differential entropy of latent representation z
f_b, f_l	Encoder networks for byte and length modalities
$\mathcal{G}_{l2b}, \mathcal{G}_{b2l}$	Cross-modal generators (length-to-byte, byte-to-length)
ϵ_θ	Noise prediction network in diffusion model
z^t	Noisy sample at time step t in diffusion process
x'	Alternative notation for diffusion process variables
w_b, w_l	Entropy-based fusion weights
λ_b, λ_l	Mixing coefficients for hybrid entropy strategy
$\mathcal{L}_{\text{class}}, \mathcal{L}_{\text{KL}}, \mathcal{L}_{\text{gen}}$	Classification, KL divergence, and generation losses
$\lambda_{\text{KL}}, \lambda_{\text{gen}}$	Loss function weighting hyperparameters

(1) Background Traffic Removal: Removing non-informative background traffic prior to detection enhances both detection efficiency and precision by reducing false positives. We identify and eliminate the following categories of background traffic:

- Failed TCP handshakes (e.g., missing responses to SYN packets or RST resets), primarily caused by scanning activities or misconfigurations, which generally lack malicious payloads;
- Routine DNS queries via UDP port 53, representing a substantial portion of benign DNS traffic and typically non-malicious in nature;
- LLMNR/NBNS protocol traffic, employed for local network name resolution, often regarded as network noise and seldom linked to adversarial activities;
- NTP and ICMP traffic, serving fundamental network maintenance functions and rarely indicative of direct attack vectors.

(2) Flow Segmentation: Traffic segmentation is commonly performed at the packet-level, flow-level, or session-level granularity. In this work, we segment raw network traffic into flows using the standard five-tuple definition: {source IP, destination IP, source port, destination port, protocol}.

We adopt flow-level segmentation as the foundational unit for our analysis, motivated by two key advantages. First, in contrast to packet-level representations – which treat each packet independently – flow-level aggregation captures directional coherence and temporal dependencies within network communications. Second, compared to session-level representations, flow-level structures are structurally simpler, as they do not require maintaining bidirectional state information.

(3) Irrelevant Field Processing: Since our model relies on semantic and behavioral features derived from raw network traffic for classification, it is essential to remove fields that are irrelevant to these characteristics. This helps prevent potential negative impacts on the training process and model generalization. Specifically, we eliminate MAC addresses and IP addresses – fields that uniquely identify devices – and replace their original values with “0x00”.

(4) Multimodal Feature Extraction: For subsequent multimodal learning, we select two complementary modalities: the raw bytes of each flow and the packet length sequence. The rationale for this selection is twofold. First, raw bytes encapsulate both protocol header information and payload content, thereby capturing the semantic characteristics of network traffic. Second, packet length sequences reflect temporal communication patterns, capturing observable behavioral attributes of the flow.

Specifically, for raw bytes, we extract the first N packets from each flow, including H bytes from the packet header and P bytes from the payload. This results in a total of $N(H + P)$ bytes per flow. Any missing data due to insufficient packet count are padded with “0x00”. Similarly, for the packet length sequence, we extract the lengths of the first M packets per flow, padding with zeros when necessary.

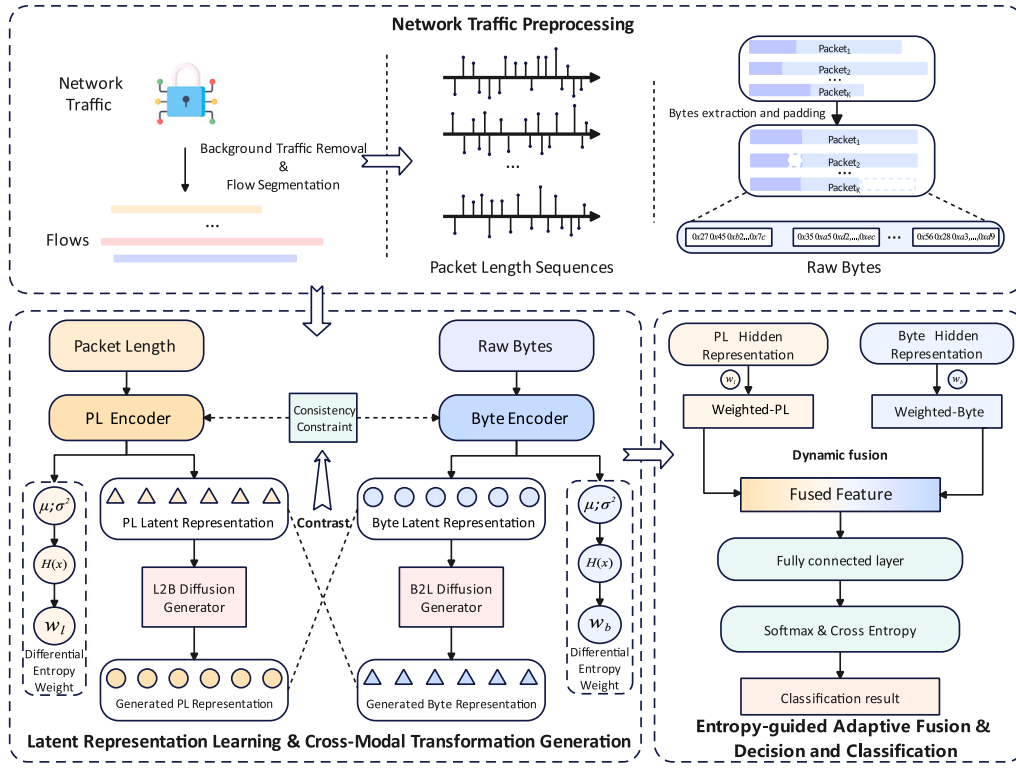


Fig. 1. Architecture overview of the ER-CMGI framework. The model consists of five key modules: (1) Network Traffic Preprocessing, (2) Latent Representation Learning, (3) Cross-Modal Transformation Generation, (4) Entropy-Guided Adaptive Fusion, and (5) Decision and Classification. Arrows indicate the flow of data and information across components.

Importantly, both modalities inherently preserve temporal information crucial for detecting attack patterns. The raw byte sequences maintain chronological packet ordering, capturing the temporal evolution of communication content, while the packet length sequences represent time-ordered size variations that reflect communication rhythms and behavioral patterns. This temporal structure enables our subsequent CNN architectures to exploit temporal dependencies: the two-dimensional CNN for byte sequences models temporal correlations across packets alongside spatial dependencies within packets, while the one-dimensional CNN for packet length sequences captures temporal dynamics such as periodicity and burstiness commonly exhibited by specific applications or attack vectors.

These strategies ensure a comprehensive characterization of both fine-grained payload patterns and high-level communication dynamics. By integrating micro-level payload details with macro-level behavioral traits, the two modalities effectively enrich the model's understanding of network traffic, thereby enhancing the effectiveness of multimodal learning.

The selection of byte sequences and packet length sequences as our primary modalities is motivated by their complementary advantages in encrypted traffic analysis. Compared to alternative combinations such as time intervals or header fields, our chosen modalities offer superior *encryption robustness* (both remain observable under strong encryption), *semantic completeness* (bytes capture protocol structures while lengths capture behavioral patterns), and *universal applicability* (extractable from any network traffic regardless of protocol type). Time intervals, while providing temporal information, suffer from network jitter and deployment variability. Header-payload combinations become ineffective in encrypted scenarios where payload content is inaccessible.

3.2. Representation learning

To effectively leverage multimodal network traffic data for deep learning tasks, it is essential to first transform raw traffic samples into

structured and semantically meaningful representations. This section presents the design of our probabilistic representation learning framework, which not only encodes heterogeneous modalities – namely byte sequences and packet length sequences – into a shared latent space but also quantifies the uncertainty associated with each modality's learned representation. These probabilistic representations serve as the foundational inputs for subsequent multimodal fusion and decision-making modules.

3.2.1. Problem formalization

Let $\mathcal{X} = \{x_b, x_l\}$ denote a multimodal network traffic sample, where $x_b \in \mathbb{R}^{S \times F}$ represents the byte sequence and $x_l \in \mathbb{R}^S$ denotes the packet length sequence, with subscripts b and l indicating byte and length modalities respectively. Here, S refers to the number of packets extracted from the flow, and F indicates the byte feature dimension.

This module aims to learn probabilistic latent representations for each modality, where the encoders parameterize Gaussian distributions $z_b \sim \mathcal{N}(\mu_b, \sigma_b^2)$ and $z_l \sim \mathcal{N}(\mu_l, \sigma_l^2)$ within a shared latent space, with (μ_b, σ_b^2) and (μ_l, σ_l^2) being neural network outputs. Additionally, the information uncertainty of each modality's latent representation is quantified using differential entropy. This provides a theoretical basis for the subsequent cross-modal fusion and decision-making mechanisms.

3.2.2. Probabilistic encoder

The probabilistic encoding mechanism is based on a VAE architecture, comprising two distinct encoders: one dedicated to byte sequences and the other to packet length sequences. Although both encoders are implemented using CNNs, they are designed to capture modality-specific characteristics.

Specifically, the byte sequence encoder employs a two-dimensional CNN to model both temporal correlations across packets and spatial dependencies within individual packets. This architecture enables the extraction of fine-grained patterns such as protocol header structures or application-layer payload features. In contrast, the packet length encoder utilizes a one-dimensional CNN to capture temporal

dynamics in packet sizes—such as periodicity, burstiness, or rhythmic communication patterns commonly exhibited by certain applications.

Both encoders consist of stacked convolutional layers followed by batch normalization and pooling operations, progressively extracting high-level abstract features. The final output of each encoder is projected into the shared latent space through fully connected layers.

As illustrated in Fig. 2, both unimodal encoders generate the mean μ and log-variance $\log \sigma^2$ of the latent vectors, which parameterize multivariate Gaussian distributions. Using the reparameterization trick, latent vectors are sampled as follows:

$$\begin{aligned} z_b &= \mu_b + \epsilon \odot \sigma_b, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \\ z_l &= \mu_l + \epsilon \odot \sigma_l, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \end{aligned} \quad (1)$$

where $\mathbf{I}$ represents the identity matrix.

This stochastic encoding strategy not only captures the distributional properties of the input data but also introduces a principled mechanism for quantifying representational uncertainty.

3.2.3. Uncertainty quantification based on differential entropy

To quantify the information uncertainty embedded in the latent representations, we leverage the concept of differential entropy from information theory. For a multivariate Gaussian distribution derived from a unimodal encoder, the differential entropy can be analytically expressed as

$$H(z) = \frac{1}{2} \sum_{i=1}^d \log(2\pi e \sigma_i^2), \quad (2)$$

where d is the latent space dimension and σ_i^2 denotes the variance of the i th dimension.

For practical computation, the average entropy per sample is simplified as

$$H(z) = \frac{1}{2} \log(2\pi e) + \frac{1}{2d} \sum_{i=1}^d \log \sigma_i^2. \quad (3)$$

To ensure numerical stability during training, we apply a clamping operation on the logarithmic variances:

$$H(z) = \frac{1}{2} \log(2\pi e) + \frac{1}{2d} \sum_{i=1}^d \text{clamp}(\log \sigma_i^2, \alpha_{\min}, \alpha_{\max}), \quad (4)$$

where $\alpha_{\min}$ and $\alpha_{\max}$ are hyperparameters defining the lower and upper bounds, respectively. The clamp function ensures that values remain within a numerically stable range:

$$\text{clamp}(x, \min, \max) = \begin{cases} \min, & \text{if } x < \min \\ \max, & \text{if } x > \max \\ x, & \text{otherwise.} \end{cases} \quad (5)$$

Differential entropy offers a theoretically grounded and computationally efficient measure of uncertainty. It allows us to distinguish between two types of uncertainty: aleatoric uncertainty, which stems from inherent randomness in the data, and epistemic uncertainty, which reflects the model's lack of knowledge or confidence.

In the context of multimodal network traffic analysis, the uncertainty associated with each modality carries distinct semantic implications. High entropy in byte sequences may indicate encrypted content or high variability in payload structures, which can serve as valuable discriminative features for certain classification tasks, particularly in distinguishing encrypted malicious traffic from normal patterns. Similarly, elevated entropy in packet length sequences may suggest irregular communication patterns or complex temporal behaviors, which could be indicative of specific application types or malicious activities rather than simply representing noise.

By jointly quantifying uncertainty across both modalities, the model can assess the information content and discriminative potential of each modality for a given traffic sample. Rather than treating high entropy as inherently unreliable, this approach recognizes that different

entropy patterns may carry complementary classification signals. The theoretical foundation enables adaptive modality fusion that dynamically weights each modality based on its discriminative contribution to the specific classification context, supporting informed decision-making throughout the system while leveraging the full spectrum of available information.

3.3. Cross-modal transformation generation

We propose a cross-modal transformation mechanism based on diffusion models, which enables bidirectional mapping between different modalities of network traffic features by replacing the decoder in the conventional VAE framework. This design aims to enhance modality complementarity and improve the quality of latent representation learning.

The adoption of diffusion models over simpler generators such as MLPs or CNNs is motivated by their strong theoretical foundation and practical advantages. Diffusion models excel at capturing complex, multimodal distributions through an iterative denoising process [39], making them well-suited for modeling the intricate relationships among heterogeneous network traffic modalities. Unlike deterministic architectures that learn direct mappings, diffusion models estimate the underlying score function of the data distribution [40], enabling more flexible and robust cross-modal transformations. The progressive denoising mechanism also provides implicit regularization, which helps prevent overfitting to specific patterns [41]—a critical property for generalization in security-sensitive applications. Furthermore, existing studies show that diffusion models achieve superior sample quality and mode coverage in conditional generation tasks [42], directly contributing to better cross-modal semantic alignment in our framework.

Unlike traditional multimodal fusion approaches that rely on direct feature concatenation or early/late fusion strategies, our method leverages diffusion models to generate the raw features of one modality from the latent representation of another. This approach not only strengthens semantic alignment across modalities but also enriches latent representations through cross-modal generation consistency constraints, thereby promoting more robust and generalizable feature learning.

3.3.1. Lightweight diffusion model design

Diffusion models generate data by iteratively denoising samples that are progressively corrupted with noise over a series of time steps. The overall architecture is illustrated in Fig. 2. To enhance computational efficiency for network traffic analysis, we design a lightweight variant incorporating the following key optimizations:

1. **Optimized Time Steps:** Instead of using hundreds to thousands of iterations as in standard diffusion models, we reduce the number of time steps to T , significantly improving inference speed while maintaining high-quality generation.
2. **Linear Noise Scheduling:** We adopt a linear noise schedule where the noise level β_t increases linearly from β_1 to β_T . This balances sampling fidelity and convergence speed.
3. **Conditional Diffusion Architecture:** We adapt the diffusion model for conditional generation, defined formally as:

$$p_\theta(\hat{z}^0 | z_c) = \int p_\theta(\hat{z}^{0:T} | z_c) d\hat{z}^{1:T}, \quad (6)$$

where $\hat{z}^0$ denotes the generated target modality features (the final clean output), z_c is the conditional modality's latent representation (either z_b or z_l), and $p_\theta(\hat{z}^{0:T} | z_c)$ is modeled via a Markov chain:

$$p_\theta(\hat{z}^{0:T} | z_c) = p(\hat{z}^T) \prod_{t=1}^T p_\theta(\hat{z}^{t-1} | \hat{z}^t, z_c). \quad (7)$$

Here, $\hat{z}^T$ represents pure noise, and the chain progressively denoises to produce the target features.

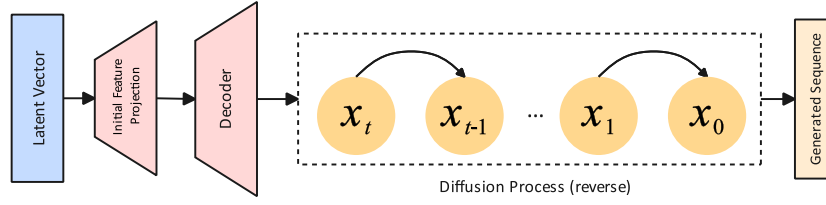


Fig. 2. The structure of the lightweight diffusion model.

4. **Network Architecture:** The core denoising network ϵ_θ employs a lightweight U-Net structure consisting of multiple down-sampling and up-sampling blocks. The conditional information z_c is integrated into the decoding process through conditional injection techniques.

3.3.2. Bidirectional cross-modal generator

To enable full bidirectional transformation between modalities, we design two complementary generative components:

1. **Byte-to-Length Generator (B2L):** Given the byte modality latent representation $z_b \in \mathbb{R}^d$ as condition, we generate the length modality features through the diffusion process. Starting from noise $\hat{z}_l^T \sim \mathcal{N}(0, \mathbf{I})$, the reverse denoising process is learned as:

$$\hat{z}_l^{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\hat{z}_l^t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\hat{z}_l^t, t, z_b) \right) + \sigma_t \xi, \quad (8)$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$, $\sigma_t^2 = \beta_t$, and $\xi \sim \mathcal{N}(0, \mathbf{I})$ is independent noise.

2. **Length-to-Byte Generator (L2B):** Conditioned on the length modality representation z_l , this generator reconstructs the corresponding byte representation through the diffusion process. Starting from noise $\hat{z}_b^T \sim \mathcal{N}(0, \mathbf{I})$, the denoising process follows:

$$\hat{z}_b^{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\hat{z}_b^t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\hat{z}_b^t, t, z_l) \right) + \sigma_t \xi. \quad (9)$$

Given that the byte modality typically exhibits a significantly higher feature dimension compared to the packet length modality, we incorporate feature expansion layers within the L2B generator. These layers progressively increase the feature dimensions through grouped convolutions and up-sampling operations, ensuring an effective mapping from low-dimensional length features to high-dimensional byte representations.

As illustrated in Fig. 3, the network adopts a two-dimensional convolutional architecture with skip connections, which helps preserve fine-grained structural details during the generation process.

Importantly, the proposed lightweight U-Net does not rely on computationally intensive attention mechanisms. Instead, it achieves efficient cross-modal alignment through carefully designed convolutional layers and hierarchical skip connections, making it particularly suitable for network traffic analysis applications.

3.3.3. Cross-modal generation consistency constraint

Cross-modal generation consistency serves as a critical mechanism to align latent representations across modalities and ensure accurate reconstruction of original data. It is formulated as:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{gen}}^b + \mathcal{L}_{\text{gen}}^l, \quad (10)$$

where

$$\begin{aligned} \mathcal{L}_{\text{gen}}^b &= \|x_b - \hat{x}_b\|_2^2, & \hat{x}_b &= \mathcal{G}_{l2b}(z_l), \\ \mathcal{L}_{\text{gen}}^l &= \|x_l - \hat{x}_l\|_2^2, & \hat{x}_l &= \mathcal{G}_{b2l}(z_b). \end{aligned} \quad (11)$$

Here, $z_b = f_b(x_b)$ and $z_l = f_l(x_l)$ are latent representations obtained from their respective encoders, and $\mathcal{G}_{l2b}$ and $\mathcal{G}_{b2l}$ denote the cross-modal

generators. Unlike traditional cycle-consistency constraints, this direct generation loss is more computationally efficient while still promoting strong alignment between modalities.

To accommodate the characteristics of each modality, we introduce an adaptive weighting strategy based on a hybrid entropy mechanism. Empirical analysis reveals that byte modality performs better with higher entropy samples, while length modality is more reliable with lower entropy samples.

Based on these observations, we derive the optimal fusion weights from information-theoretic principles. Let the classification performance of modality m be correlated with its entropy $H(z_m)$. The optimal weight should reflect both the entropy-performance relationship and the modality's information content. We formulate this as:

$$w_m = \lambda_m \cdot \underbrace{\frac{H(z_m)}{\sum_j H(z_j)}}_{\text{High-entropy priority}} + (1 - \lambda_m) \cdot \underbrace{\frac{1/H(z_m)}{\sum_j 1/H(z_j)}}_{\text{Low-entropy priority}} \quad (12)$$

where λ_m controls the balance between the two strategies based on the modality-specific entropy-accuracy relationship. For byte modality, $\lambda_b = 0.7$ (favoring high entropy); for length modality, $\lambda_l = 0.2$ (favoring low entropy). The practical implementation uses:

$$\begin{aligned} w_b &= \lambda_b \cdot \frac{1}{\bar{H}(z_b) + \epsilon} + (1 - \lambda_b) \cdot \frac{\bar{H}(z_b)}{\bar{H}(z_b) + \bar{H}(z_l)}, \\ w_l &= \lambda_l \cdot \frac{1}{\bar{H}(z_l) + \epsilon} + (1 - \lambda_l) \cdot \frac{\bar{H}(z_l)}{\bar{H}(z_b) + \bar{H}(z_l)}, \end{aligned} \quad (13)$$

where $\bar{H}(z)$ represents the sample-averaged differential entropy of modality z (computed as the average of $H(z)$ across a batch of samples), λ_b and λ_l are blending coefficients that control the balance between inverse entropy weighting (favoring high uncertainty) and standard entropy weighting (favoring low uncertainty), and ϵ is a small constant for numerical stability.

Finally, the weights are normalized to ensure they sum to one:

$$w'_b = \frac{w_b}{w_b + w_l}, \quad w'_l = \frac{w_l}{w_b + w_l}. \quad (14)$$

Accordingly, the weighted generation loss becomes:

$$\mathcal{L}_{\text{gen}} = w'_b \cdot \mathcal{L}_{\text{gen}}^b + w'_l \cdot \mathcal{L}_{\text{gen}}^l. \quad (15)$$

This hybrid entropy-based adaptive weighting strategy enables the model to flexibly adjust the relative importance of each modality within the generation consistency constraint, thereby enhancing both the accuracy and robustness of cross-modal transformation.

3.4. Entropy-guided adaptive fusion

This section introduces a dynamic feature fusion strategy grounded in information theory. By leveraging the differential entropy of latent representations, this module guides the optimal integration of multimodal features, achieving adaptive decision optimization.

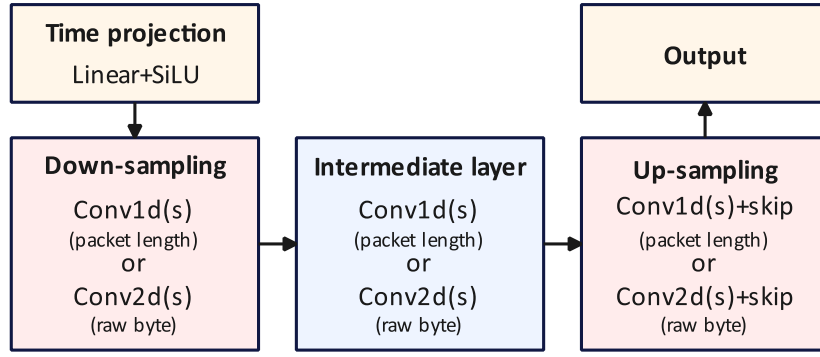


Fig. 3. The structure of the lightweight U-Net used in the diffusion model.

3.5. Entropy-guided adaptive fusion

This section introduces a dynamic feature fusion strategy grounded in information theory. By leveraging the differential entropy of latent representations, this module guides the optimal integration of multimodal features, achieving adaptive decision optimization.

Unlike traditional multimodal learning approaches that typically rely on simple concatenation or element-wise averaging of features, we propose an entropy-guided hierarchical fusion architecture. In the first stage, modality-specific weights are computed using the hybrid entropy weighting strategy that combines both traditional low-entropy priority and inverse entropy weighting through learnable mixing coefficients. These weighted latent representations are then fused through a nonlinear mapping network defined as:

$$z_{\text{fused}} = F([w'_b \cdot z_b; w'_l \cdot z_l]), \quad (16)$$

where F denotes a multilayer perceptron (MLP) with two hidden layers of dimension h , each followed by a ReLU activation function and Dropout regularization. Specifically,

$$F(z) = W_2 \cdot \text{Dropout}(\text{ReLU}(W_1 \cdot z + b_1)) + b_2. \quad (17)$$

This design enables the model to learn complex, nonlinear interactions between modalities, thereby significantly enhancing its representational capacity compared to conventional linear fusion methods.

The entropy-guided fusion weights are dynamically computed for each input sample based on the information content of individual modalities. This adaptive mechanism automatically adjusts the contribution of each modality according to its reliability and discriminative power, eliminating the need for manual parameter tuning. Compared to fixed-weight schemes or simple concatenation-based fusion methods, this approach provides dynamic adaptability to diverse traffic patterns and improved robustness when individual modalities are corrupted by noise or missing information.

3.6. Decision and classification

The decision-making component takes as input the fused representation z_{fused} , which is generated by the entropy-guided fusion module, and produces the final classification output through a linear mapping:

$$p(y | z_{\text{fused}}) = \text{softmax}(W \cdot z_{\text{fused}} + b). \quad (18)$$

It is important to note that the fused representation z_{fused} has already undergone multiple nonlinear transformations and Dropout regularization within the fusion layers. As a result, we adopt a simple single-layer classifier to avoid over-parameterization and maintain model efficiency, without compromising performance.

To train this classification framework effectively, we define a comprehensive loss function that jointly optimizes three key objectives to ensure both classification accuracy and representation quality:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{class}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{gen}} \mathcal{L}_{\text{gen}}. \quad (19)$$

The first component is the classification loss, which serves as the primary task objective and is computed using cross-entropy:

$$\mathcal{L}_{\text{class}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c}), \quad (20)$$

where $y_{i,c}$ is the ground truth label and $p_{i,c}$ is the predicted probability for class c of sample i .

The second component provides latent space regularization through KL divergence, which encourages the latent representations of both modalities to follow standard normal distributions:

$$\mathcal{L}_{\text{KL}} = \text{KL}(q(z_b|x_b)||p(z)) + \text{KL}(q(z_l|x_l)||p(z)), \quad (21)$$

where $p(z) = \mathcal{N}(0, I)$ is the prior. This constraint improves the structure and generalization ability of the latent space, reducing the risk of overfitting to specific patterns.

The third component enforces cross-modal generation consistency to ensure semantic alignment between different modalities. This loss is defined as:

$$\mathcal{L}_{\text{gen}} = \frac{1}{N} \sum_{i=1}^N [\|x_{b,i} - \hat{x}_{b,i}\|_2^2 + \|x_{l,i} - \hat{x}_{l,i}\|_2^2], \quad (22)$$

where $\hat{x}_{b,i} = \mathcal{G}_{l2b}(z_{l,i})$ and $\hat{x}_{l,i} = \mathcal{G}_{b2l}(z_{b,i})$ denote the byte sequences generated from the length latent representation and the packet length sequences generated from the byte latent representation, respectively. Here, $z_{b,i} = f_b(x_{b,i})$ and $z_{l,i} = f_l(x_{l,i})$ are the latent representations obtained from their respective encoders.

This design ensures that latent representations from one modality can accurately reconstruct the other modality, cross-modal semantic alignment is maintained throughout the learning process, and the lightweight diffusion generators effectively capture inter-modal relationships. The cross-modal generation consistency approach enables self-supervised learning through direct cross-modal generation and comparison, enhancing the model's ability to learn meaningful representations without additional labeled data.

The combination of these three loss components creates a balanced training objective that promotes both discriminative classification performance and robust multimodal representation learning, ensuring the model's effectiveness across diverse network security scenarios.

3.7. Theoretical foundation

Although our model does not incorporate the traditional self-modal reconstruction loss commonly used in VAEs, we provide a theoretical justification demonstrating that the combination of cross-modal generation consistency loss and KL divergence establishes an equivalent formulation to the Evidence Lower Bound (ELBO) in a multimodal setting.

We establish a unified theoretical framework based on the following objective function:

$$\mathcal{L}_{\text{unified}} = \mathcal{L}_{\text{gen}} + \lambda \mathcal{L}_{\text{KL}}. \quad (23)$$

Under appropriate conditions, minimizing this objective is mathematically equivalent to maximizing the multimodal variational lower bound:

$$\mathcal{L}_{\text{ELBO-joint}} = \mathbb{E}[\log p(x_l|z_b) + \log p(x_b|z_l)] - \mathcal{L}_{\text{KL}}. \quad (24)$$

This equivalence relies on the assumption that the cross-modal generation functions in our lightweight diffusion model are sufficiently expressive. When the generators $\mathcal{G}_{l2b}$ and $\mathcal{G}_{b2l}$ have adequate representational capacity, the generation consistency losses can effectively approximate the cross-modal log-likelihoods.

The key insight underlying our approach is that cross-modal generation consistency enforces a stronger constraint than traditional self-reconstruction. Specifically, our framework requires that latent representations from one modality must contain sufficient semantic information to accurately reconstruct data from another modality:

$$\mathcal{G}_{l2b}(f_l(x_l)) \approx x_b \quad \text{and} \quad \mathcal{G}_{b2l}(f_b(x_b)) \approx x_l, \quad (25)$$

where $f_b(\cdot)$ and $f_l(\cdot)$ denote the byte and length encoders, respectively.

This cross-modal constraint provides three main theoretical advantages. First, it requires the latent space to capture shared semantic concepts across modalities rather than modality-specific information. Second, it acts as an implicit regularizer that prevents encoders from learning artifacts or noise. Third, combined with KL divergence terms, it facilitates alignment between different modal latent representations through a shared prior distribution $p(z)$.

From an information-theoretic perspective, our formulation ensures that the learned representations satisfy the following property:

$$I(Z_b; Z_l) \geq \alpha, \quad (26)$$

where $I(\cdot; \cdot)$ denotes mutual information, and $\alpha > 0$ is a positive constant. This guarantees that the latent representations from different modalities share sufficient common information, establishing strong cross-modal semantic alignment while preserving modality-specific characteristics.

Therefore, our framework preserves the theoretical foundation of variational inference while introducing a self-supervised learning mechanism through cross-modal generation constraints. This approach compensates for the absence of traditional self-modal reconstruction loss and provides theoretical guarantees for multimodal representation learning.

4. Experiment

4.1. Experimental evaluation

A. Datasets.

We evaluate our ER-CMGI model on two publicly available datasets:

D1: The complete USTC-TFC2016 dataset [43] containing 20 traffic categories (10 normal and 10 malicious). This comprehensive collection includes application-layer network traffic for malware analysis and intrusion detection, covering common protocols (HTTP, FTP, SMTP) and malicious activities (Cridex, Geodo, Zeus).

D2: A composite dataset combining 5 selected malware families from Malicious_TLS [44] with 5 selected application categories from CrossNet2021 [19]. The malicious component consists of TLS traffic collected from real networks (2018–2021), while the benign component includes applications tested under realistic network conditions with packet loss and latency.

B. Experimental setup.

Our implementation utilizes PyTorch 2.5.1 and Python 3.10.14 on Ubuntu 22.04, equipped with an NVIDIA V100 (32 GB) GPU and an Intel Xeon Gold 5218R CPU. We evaluate performance using standard metrics: Accuracy, Recall, Precision, and F1-score.

Key implementation parameters are shown in Table 3. We use the Adam optimizer with a learning rate of 10^{-4} and weight decay of

Table 3

Implementation details and hyperparameter settings.

Parameter	Value	Parameter	Value
Packets per flow	20	Header size	54 bytes
Payload size	256 bytes	Length features	5-dim
Latent dimension	64	Hidden dimension	128
Diffusion steps	10	UNet hidden dim	32
Time embedding dim	32	Convolution groups	8
Optimizer	Adam	Learning rate	10^{-4}
Weight decay	5×10^{-4}	Batch size	32
LR scheduler	ReduceLROnPlateau	LR reduction factor	0.5
Early stop patience	10 epochs	Gradient clipping	5.0
λ_{KL}	0.05	λ_{gen}	0.1
Byte entropy weight	0.7	Length entropy weight	0.2

5×10^{-4} . The learning rate scheduling employs ReduceLROnPlateau with a reduction factor of 0.5 and patience of 5 epochs. Early stopping is applied with a patience of 10 epochs, and gradient clipping is set to 5.0 for training stability. A batch size of 32 was selected based on preliminary experiments balancing training efficiency, memory usage, and model performance. Loss function weights are set to $\lambda_{\text{KL}} = 0.05$ and $\lambda_{\text{gen}} = 0.1$, while the hybrid entropy strategy uses mixing coefficients of 0.7 for raw byte modality and 0.2 for packet length sequence modality.

The lightweight diffusion model incorporates several architectural optimizations to reduce computational overhead while maintaining generation quality. The diffusion timesteps are reduced from the standard 1000 to 10 steps using a cosine_simple noise scheduling strategy. The UNet noise predictor features a hidden dimension of 32 and time embedding dimension of 32, significantly smaller than standard implementations. The architecture employs grouped convolutions (groups = 8) combined with pointwise convolutions to reduce parameter count, while using GroupNorm instead of BatchNorm for improved efficiency. For byte sequence generation, we adopt a depthwise separable convolution structure; for packet length sequence generation, we use 1D convolutions with residual connections. All activation functions use SiLU (Swish) to enhance training stability and convergence speed.

C. Benchmark models.

We compare against 9 existing approaches spanning unimodal and multimodal learning paradigms:

Unimodal Models:

- **1D-CNN** [45]: An end-to-end encrypted traffic classification framework based on one-dimensional convolutional neural networks (1D-CNNs), which integrates raw input processing, feature extraction, and classification into a unified architecture to automatically learn nonlinear relationships between input and output.
- **ACID** [46]: A lightweight intrusion detection system that leverages low-dimensional embeddings learned by a compact neural model with adaptive clustering, aiming to improve sensitivity to subtle variations in traffic patterns through enhanced inter-class separation.
- **IIT** [21]: A heterogeneous encrypted traffic classification model employing dual multi-head self-attention encoders to jointly model intra-flow packet dynamics and inter-flow traffic dependencies, thereby capturing both temporal and relational characteristics of network behavior from complementary perspectives.
- **RFG-HELAD** [47]: A robust fine-grained attack detection framework that combines contrastive learning, GAN-based adversarial defense, and Deep kNN-based out-of-distribution detection to address the open-set problem and enhance resilience against adversarial perturbations.
- **FastTraffic** [48]: A lightweight deep learning method for encrypted traffic classification that treats packets as text-like sequences, utilizes N-gram feature embedding to capture structured and sequential patterns, and employs a compact three-layer MLP for fast and resource-efficient inference.

Table 4
Comparison results on the D1(%).

	Method	Accuracy	Recall	Precision	F1
Unimodal	1D CNN [45]	89.76	90.37	90.32	90.25
	ACID [46]	94.13	93.92	94.19	94.53
	IIT [21]	97.86	97.15	97.55	97.68
	RFG-HELAD [47]	85.18	84.76	84.98	82.68
	FastTraffic [48]	96.93	94.97	96.57	95.51
Multimodal	AppNet [30]	91.31	91.54	91.24	90.87
	MIMETIC [14]	93.25	93.14	93.42	93.07
	DISTILLER [15]	95.23	95.13	95.21	95.24
	AN-Net [31]	98.54	98.57	98.67	98.42
	ER-CMGI	99.12	99.17	99.23	99.12

Multimodal Models:

- **AppNet [30]:** An end-to-end hybrid CNN-RNN architecture designed to jointly model sequential flow patterns and application-specific signatures from raw TLS traffic for mobile app identification via parallel feature extraction pathways.
- **MIMETIC [14]:** A multimodal deep learning framework for encrypted mobile traffic classification that captures both intra- and inter-modality dependencies from heterogeneous data sources, enabling more effective modeling of complex traffic patterns through multi-view representation learning.
- **DISTILLER [15]:** A multitask multimodal deep learning architecture that jointly learns intra- and inter-modality relationships across diverse traffic features, supporting simultaneous optimization over multiple classification objectives within a unified model.
- **AN-Net [31]:** An anti-noise deep learning architecture tailored for anonymous traffic classification, featuring robust short-term feature extraction and an enhanced multimodal fusion mechanism to mitigate both packet-level noise and attribute perturbations.

To ensure fair and valid comparisons across all methods, all benchmark models are trained and evaluated under identical conditions: (1) *Consistent Data Preprocessing*: All methods apply the same preprocessing pipeline detailed in Section 3.1, including background traffic removal, flow segmentation, and feature extraction procedures. (2) *Uniform Dataset Partitioning*: We use a consistent 70%–10%–20% split for training, validation, and testing across all experiments, with stratified sampling to maintain class distribution balance. (3) *Standardized Training Environment*: All models are trained on the same hardware configuration with identical computational resources, using PyTorch's deterministic settings (fixed random seeds) to ensure reproducible results. (4) *Consistent Evaluation Metrics*: Performance evaluation employs the same metrics (Accuracy, Recall, Precision, F1-score) with identical calculation methods. (5) *Hyperparameter Optimization*: For each baseline method, we conduct grid search or follow the original paper's recommendations to ensure optimal performance, while maintaining consistency in optimization procedures across all comparisons.

4.2. Comparison with benchmarks

The comparison results of ER-CMGI and benchmark on two datasets are shown in Tables 4 and 5.

The experimental results demonstrate that our proposed ER-CMGI model significantly outperforms both unimodal and multimodal baseline methods on two datasets. The performance gap between ER-CMGI and traditional unimodal approaches is particularly notable. Compared to the best unimodal method IIT, ER-CMGI improves accuracy by 1.26% and 1.47% on D1 and D2, respectively. In comparison to FastTraffic, the accuracy improvements reach 1.81% on D1 and 5.38% on D2, demonstrating the inherent limitations of unimodal methods in capturing complex patterns in encrypted traffic.

Table 5
Comparison results on the D2(%).

	Method	Accuracy	Recall	Precision	F1
Unimodal	1D CNN [45]	87.86	88.14	88.09	87.76
	ACID [46]	92.87	92.64	92.35	92.79
	IIT [21]	96.36	96.12	96.17	96.06
	RFG-HELAD [47]	88.29	87.63	87.94	88.26
	FastTraffic [48]	92.16	91.52	91.33	91.83
Multimodal	AppNet [30]	90.32	90.24	90.29	90.31
	MIMETIC [14]	91.87	91.74	91.82	91.84
	DISTILLER [15]	94.68	94.63	94.57	94.52
	AN-Net [31]	97.19	97.32	97.29	97.31
	ER-CMGI	97.83	97.75	97.92	97.81

When compared with existing multimodal frameworks, ER-CMGI still exhibits clear advantages. Relative to the closest competitor AN-Net, our model achieves 0.58% and 0.64% accuracy improvements on D1 and D2, respectively. Compared to other multimodal methods such as DISTILLER and MIMETIC, ER-CMGI demonstrates accuracy improvements of 5.87% and 3.89% on D1, and 5.96% and 3.15% on D2. This performance gain stems from our entropy-regulated fusion mechanism and cross-modal semantic alignment, which effectively address the semantic inconsistency issues that limit other multimodal approaches.

Notably, ER-CMGI demonstrates robust performance across two datasets with distinct characteristics, validating its capability to handle diverse traffic patterns. Particularly on dataset D2, which contains malware families collected from 2018 to 2021, ER-CMGI maintains strong performance, indicating its effectiveness in addressing evolving cyber threats - a critical requirement for practical application scenarios.

The normalized confusion matrices on the two datasets are shown in Fig. 4. The results indicate that our model performs excellently in these two multi-class classification tasks. From the confusion matrices, it can be observed that the classification accuracy of most categories reaches or approaches 1. The diagonal elements generally appear dark blue, which demonstrates that the model has extremely high classification accuracy. For example, in the first dataset, applications such as FTP, WorldOfWarcraft, BitTorrent, and MySQL, as well as platforms like aiqiyi, shifu, and upatre in the second dataset, all achieve or approach a recognition accuracy of 1.

It is worth noting that even among applications with similar functions or overlapping features, the model still maintains a strong discriminative ability. In a few cases, there are slight confusion phenomena, such as between Outlook and Gmail (approximately 2%) and between Neris and Virut (approximately 11%). These confusions reflect genuine network behavior similarities rather than dataset annotation issues. The Outlook-Gmail confusion stems from both applications using identical IMAP/SMTP protocols and TLS encryption patterns, resulting in nearly indistinguishable encrypted traffic characteristics. Similarly, the Neris-Virut confusion reflects the fact that both botnet families utilize similar HTTP-based C&C communication patterns with comparable beacon intervals and payload structures. These patterns represent inherent classification challenges in encrypted traffic analysis, where functionally similar applications naturally exhibit comparable network behaviors.

Based on comprehensive analysis of the confusion matrices, the types of attacks most prone to misclassification fall into two main categories: (1) *Protocol-similar traffic*, where different applications adopt identical communication protocols, making their encrypted patterns nearly indistinguishable; and (2) *Behavioral-similar malware*, where sophisticated attacks intentionally mimic legitimate application behaviors or exhibit similar communication patterns. The underlying technical causes include protocol convergence in encrypted environments and advanced evasion techniques.

The results validate our hypothesis that traditional static fusion paradigms limit multimodal synergy, while explicit cross-modal generation constraints enhance semantic coherence. ER-CMGI's information-theoretic optimization framework successfully bridges the modality gap between statistical features and raw payloads, establishing new state-of-the-art performance for encrypted traffic analysis.

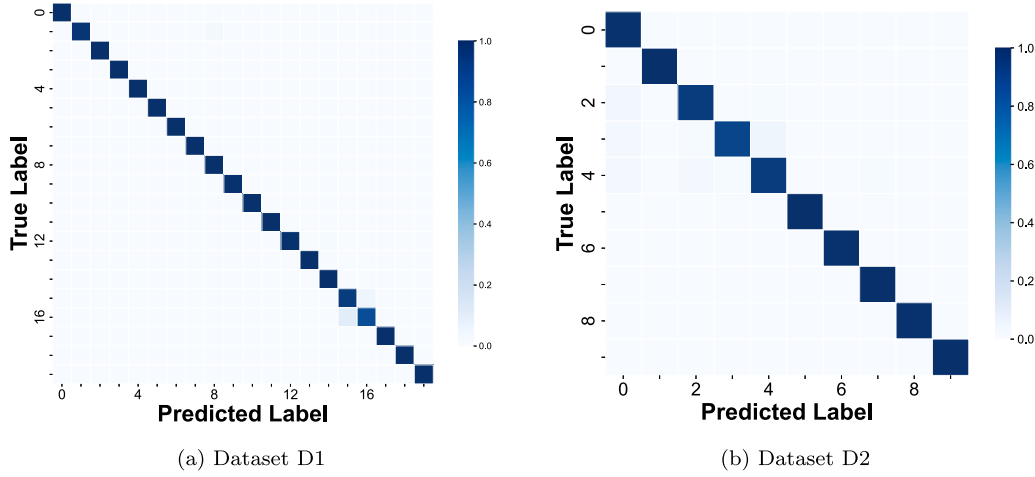


Fig. 4. Confusion matrices on two datasets.

Table 6
The ablation results on D1(%).

Model	Accuracy	Recall	Precision	F1
w/o Packet length	93.86	93.74	93.56	93.71
w/o Raw byte	91.17	91.23	90.97	91.14
w/o Entropy-guided fusion	94.17	94.22	94.19	94.15
w/o Cross-modal generation	94.87	94.76	94.92	94.83
ER-CMGI	99.12	99.17	99.23	99.12

Table 7
The ablation results on D2(%).

Model	Accuracy	Recall	Precision	F1
w/o Packet length	92.85	92.81	92.56	92.79
w/o Raw byte	90.41	90.49	90.47	90.51
w/o Entropy-guided fusion	93.23	93.19	93.08	93.21
w/o Cross-modal generation	94.27	94.25	94.14	93.89
ER-CMGI	97.83	97.75	97.92	97.81

4.3. Ablation experiment

In order to verify the effectiveness of each module of ER-CMGI, we conducted ablation experiments, mainly including four aspects: (1) removing the packet length sequence modality and only using the original byte modality; (2) removing the original byte mode and only use the packet length sequence mode; (3) removing entropy guided adaptive fusion; (4) removing cross modal generation. In the first two cases, the model actually degenerates into a uni-modal model. The experimental results are shown in Tables 6 and 7.

When the raw byte modality was removed (w/o raw byte), model performance significantly declined, with accuracy dropping to 91.12% and 90.41% on D1 and D2 datasets respectively, representing decreases of 7.95% and 7.42% compared to the full model. F1 scores also showed substantial declines, reaching 91.14% and 90.51%. This indicates that raw bytes contain critical feature information in encrypted traffic, capable of capturing fine-grained patterns in packet content and providing rich classification evidence even in encrypted states. Among all ablated components, removing the raw byte modality caused the most significant performance drop, confirming its role as the foundational pillar of the model.

When removing the packet length sequence modality (w/o packet length), model accuracy on D1 and D2 decreased to 93.86% and 92.85%, representing 5.26% and 4.98% reductions compared to the full model. Though smaller than the raw byte removal impact, this decrease remains substantial, demonstrating that packet length sequences provide crucial supplementary information. Packet length sequences reflect temporal dynamics and communication patterns in

network traffic, particularly critical for malware relying on specific communication rhythms. These results validate the necessity of our multimodal approach, which simultaneously leverages content and behavioral features for classification.

After removing the entropy-guided adaptive fusion mechanism (w/o entropy-guided adaptive fusion), accuracy on D1 and D2 decreased to 94.17% and 93.23%, representing 4.95% and 4.60% drops compared to the full model. This demonstrates that our entropy-based dynamic fusion strategy significantly outperforms traditional static fusion methods. The mechanism quantifies information entropy across modalities to dynamically adjust their decision-making weights, enabling adaptive focus on discriminative features. This flexibility proves crucial when handling complex, variable encrypted traffic with differing feature manifestations across modalities.

Removing the cross-modal generation component (w/o cross-modal generation) resulted in accuracy reductions to 94.87% and 94.27% on D1 and D2, representing 4.25% and 3.56% decreases. Though the smallest decline among the four components, this remains significant. It confirms that our lightweight diffusion-based cross-modal generation framework effectively enhances semantic consistency and complementarity between modalities. By generating mappings between modalities, the framework establishes tighter cross-modal correlations, enriching feature representations while maintaining coherence. This mechanism particularly benefits scenarios with modal imbalance, where certain encryption techniques may disrupt raw byte features but minimally affect packet length sequences.

Notably, clear synergistic effects exist between components. The complete ER-CMGI model significantly outperforms any ablated variant, with advantages exceeding simple additive contributions from individual components. For example, the entropy-guided fusion mechanism better balances contributions from raw bytes and packet length sequences, while cross-modal generation further strengthens their complementarity. This mutual reinforcement collectively enhances overall model performance.

In summary, these ablation experiments comprehensively validate the necessity and effectiveness of ER-CMGI's core components, particularly demonstrating the value of multimodal fusion, entropy-based dynamic weighting, and cross-modal semantic alignment in encrypted traffic classification.

4.4. Sensitivity analysis

In this section, we conduct a comprehensive analysis of key hyperparameters that significantly impact model performance and computational efficiency. We focus on two critical aspects: (1) the number of diffusion time steps (T) in the lightweight diffusion model, which

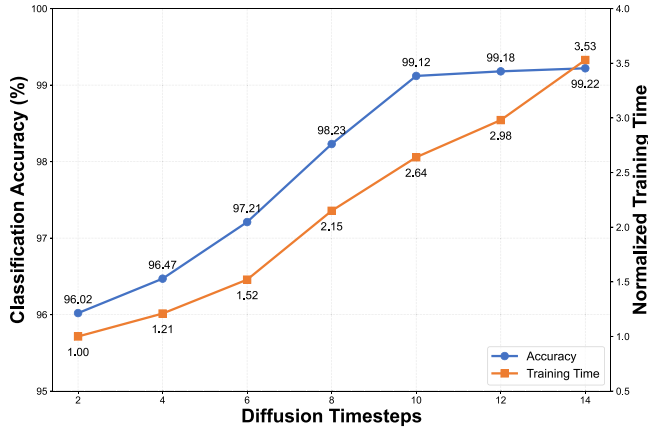


Fig. 5. Impact of diffusion timesteps on accuracy and training time.

determines the granularity of the diffusion process and affects both representational capacity and computational complexity; and (2) the weight balance among the three loss function components of formula (18), which governs the relative importance of classification, regularization, and cross-modal consistency objectives.

4.4.1. Diffusion time steps analysis

To systematically evaluate the impact of T , we conduct a series of controlled experiments, increasing the value of T from 2 to 14, while measuring the classification accuracy and the normalized training time. The normalized training time is based on the training time when $T = 2$ (set as 1.0), and the training time of other configurations is normalized relative to this benchmark to intuitively show the relative increase in computational overhead. Fig. 5 shows the results on dataset D1.

It can be observed that as the number of diffusion time steps increases, the classification accuracy of the model shows a continuous upward trend, increasing from 96.02% when $T = 2$ to 99.22% when $T = 14$. This indicates that a more fine-grained diffusion process can indeed generate more accurate feature representations, thereby improving the classification performance.

However, the improvement in classification accuracy is accompanied by a significant increase in the normalized training time. When $T = 2$, the normalized training time is 1.0, and when T increases to 14, the normalized training time rises to approximately 3.53, representing a 253% increase. It is worth noting that the curve of the improvement in classification accuracy exhibits a distinct characteristic of diminishing marginal returns. When T increases from 2 to 10, the accuracy improves by 3.10%, and the normalized training time increases to approximately 2.64. When T further increases from 10 to 14, the accuracy only improves by an additional 0.10%, while the normalized training time increases by approximately 33.7%.

Based on the above analysis, we consider that $T = 10$ is an ideal choice for balancing classification performance and computational efficiency. Under this configuration, the model achieves a high classification accuracy of 99.12%. Although the computational overhead increases by 164% compared to the baseline, the performance improvement brought about by further increasing the number of diffusion time steps is relatively limited and is difficult to offset the significant increase in computational cost. In contrast, reducing the number of time steps can lower the computational overhead but will significantly affect the classification performance. Therefore, in the experiment, we adopt $T = 10$ as the default configuration.

4.4.2. Loss weight balance analysis

To systematically evaluate the impact of loss function weight balance, we conduct a series of controlled experiments with two phases:

(1) fixing the cross-modal generation weight at $\lambda_{gen} = 0.1$ while varying the KL divergence weight λ_{KL} from 0.001 to 0.2, and (2) fixing the KL divergence weight at $\lambda_{KL} = 0.05$ while varying the cross-modal generation weight λ_{gen} from 0.01 to 0.25. Throughout all experiments, the classification weight remains fixed at $\lambda_{cls} = 1.0$ to serve as the baseline. Fig. 6 shows the results of both experimental phases on dataset D1.

It can be clearly observed from Fig. 6 that both loss components exhibit distinct inverted-U shaped performance curves, indicating optimal weight ranges for maximum classification effectiveness. For the KL divergence weight analysis, the classification accuracy increases from 97.63% when $\lambda_{KL} = 0.001$ to a peak of 99.12% when $\lambda_{KL} = 0.05$, representing an improvement of 1.49 percentage points. However, further increasing λ_{KL} to 0.2 causes the accuracy to decline significantly to 97.23%, a drop of 1.89 percentage points from the peak performance. Similarly, the cross-modal generation weight demonstrates comparable sensitivity patterns. When λ_{gen} increases from 0.01 to 0.1, the accuracy improves substantially from 97.29% to 99.12%, achieving a 1.83 percentage point gain. Beyond this optimal point, increasing λ_{gen} to 0.25 results in performance degradation to 97.59%, representing a 1.53% decrease from the peak.

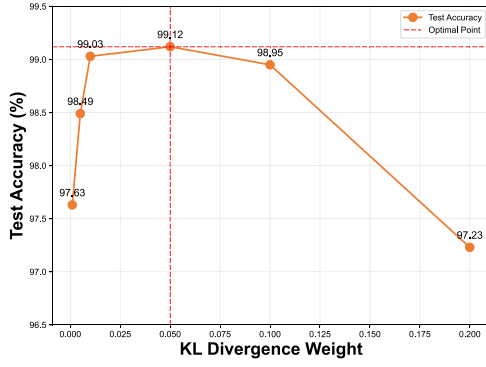
The performance curves reveal distinct characteristics of diminishing and then negative returns beyond the optimal points. For KL divergence weight, the most significant improvement occurs in the range $\lambda_{KL} \in [0.001, 0.05]$, where accuracy increases by 1.49 percentage points. When $[0.001, 0.05]$ increases from 0.05 to 0.1, the performance shows only a marginal decline of 0.17 percentage points (from 99.12% to 98.95%), but further increases lead to more substantial degradation. For cross-modal generation weight, the steepest improvement occurs in the range $\lambda_{gen} \in [0.01, 0.1]$, with an improvement of 1.83 percentage points, while increases beyond 0.1 result in consistent performance decline.

Based on the above analysis, we identify the optimal weight configuration as $\lambda_{cls} : \lambda_{KL} : \lambda_{gen} = 1.0 : 0.05 : 0.1$, which achieves the highest classification accuracy of 99.12% in both experimental phases. This configuration demonstrates that effective multimodal learning requires careful balance among the three loss components: sufficient classification guidance, appropriate latent space regularization, and optimal cross-modal consistency enforcement. Deviating from this balance either under-utilizes the auxiliary information (when weights are too small) or introduces excessive noise that interferes with the primary classification objective (when weights are too large). Therefore, in all subsequent experiments, we adopt this optimal weight configuration as the default setting.

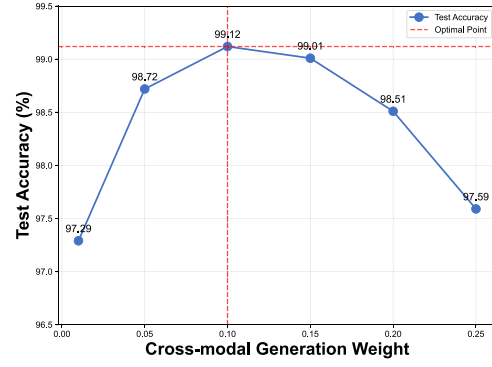
4.5. Entropy strategy analysis

This section presents an analysis of the model's core mechanism, namely the mixed entropy strategy, and its effectiveness. We perform a detailed differential entropy analysis of the ER-CMGI model on dataset D1, revealing important insights into multimodal encrypted traffic classification, as shown in Fig. 7. Specifically, we observe that the average entropy of byte sequences (0.0735) is significantly lower than that of packet length sequences (0.8775). This indicates that byte sequences in encrypted traffic tend to carry more certain and concentrated information, whereas packet length sequences exhibit higher uncertainty.

More importantly, by analyzing the entropy distributions between correctly and incorrectly classified samples, we uncover distinct patterns across modalities. In the byte modality, correctly predicted samples exhibit higher average entropy compared to misclassified ones. In contrast, in the length modality, the trend is reversed—incorrectly predicted samples show higher entropy on average. These findings suggest fundamental differences in how each modality contributes to classification performance and imply that traditional fusion strategies – such



(a) Effect of KL Divergence Weight



(b) Effect of Cross-modal Generation Weight

Fig. 6. Impact of loss function weights on classification accuracy.

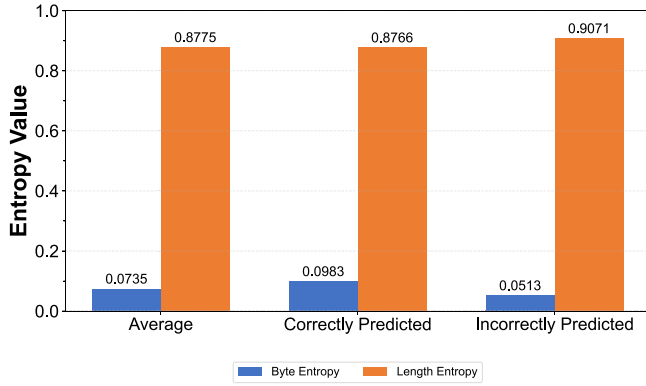


Fig. 7. Differential entropy analysis results.

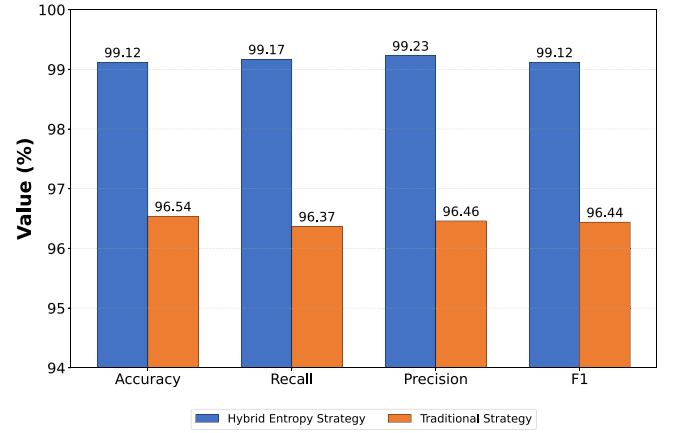


Fig. 8. Performance comparison: Hybrid vs. traditional entropy strategy.

as “low entropy, high weight” – fail to fully exploit the discriminative characteristics of individual modalities.

Motivated by the design principles established in Section 3.3.3, we propose a novel hybrid entropy-based fusion strategy. The core idea lies in introducing learnable or tunable mixing coefficients that separately control the balance between two fusion principles: “high entropy, high weight” and “low entropy, high weight” for each modality. Through systematic experimentation, we determine the optimal mixing coefficient for the byte modality to be 0.7, favoring a “high entropy, high weight” approach, while for the length modality, the optimal value is 0.2, aligning with the conventional “low entropy, high weight” paradigm. This asymmetric weighting mechanism is well-aligned with the observed entropy characteristics of the two modalities.

The effectiveness of the proposed hybrid entropy strategy is comprehensively validated through experiments, as shown in Fig. 8. Compared to the traditional entropy-weighting method, our strategy achieves improvements in accuracy (2.58%), recall (2.80%), F1 score (2.68%), and precision (2.77%). Notably, the significant improvement in F1 score highlights not only enhanced overall performance but also better class-level balance, which is crucial in real-world scenarios where class imbalance is common.

The underlying mechanism of the hybrid entropy strategy can be interpreted as a dynamically adaptive weight allocation process that respects the intrinsic statistical properties of each modality. For byte sequences, greater importance is assigned to samples with higher entropy—reflecting their potential informativeness. Conversely, for packet length sequences, higher weights are given to samples with lower entropy, prioritizing those that are more predictable. This modality-specific, sample-level adaptation enables the model to flexibly adjust its reliance on different modalities, thereby maximizing the benefits of multimodal fusion.

Experimental results demonstrate that the hybrid entropy strategy not only achieves superior performance on our dataset but also exhibits strong generalization capabilities. As such, it offers a novel and effective feature fusion approach for encrypted traffic classification and other multimodal learning tasks, opening new directions for entropy-aware design in deep multimodal modeling.

4.6. Computational complexity and efficiency analysis

To address concerns regarding computational overhead and practical deployability, we present a comprehensive analysis of the model’s complexity, efficiency, and resource requirements based on the actual architectural implementations. This evaluation directly compares our lightweight diffusion framework with standard diffusion models, demonstrating the practical feasibility and deployment advantages of the proposed approach.

We begin by analyzing the component-wise complexity and parameter distribution, as summarized in Table 8. The analysis is conducted on the real model structures, with the lightweight version containing 16.63M parameters and the standard diffusion baseline comprising 64.47M parameters.

The results show that the byte encoder is the most parameter-heavy component, accounting for 77.3% of the total due to its large fully connected layer. Meanwhile, cross-modal generators contribute 20.3% of the overall parameters. During inference, only 13.24M parameters are actively engaged (79.6% of the total), as the diffusion time embedding and noise prediction modules remain operational to ensure generation consistency verification. This selective activation

Table 8
Detailed model complexity and parameter analysis.

Component	Time complexity	Space complexity	Params (M)	Percentage
Byte encoder (2D CNN)	$O(1.6 \times 10^6)$	$O(10^5)$	12.86	77.3%
Length encoder (1D CNN)	$O(2.0 \times 10^4)$	$O(10^3)$	0.17	1.0%
Cross-modal generators	$O(2.0 \times 10^6)$	$O(10^6)$	3.37	20.3%
Entropy-guided fusion	$O(8.2 \times 10^3)$	$O(10^2)$	0.21	1.3%
Classification head	$O(1.3 \times 10^3)$	$O(10^1)$	0.002	0.01%
Total (Training)	$O(2.0 \times 10^6)$	$O(10^6)$	16.63	100%
Total (Inference)	$O(1.6 \times 10^6)$	$O(10^5)$	13.24	79.6%

Table 9
Framework efficiency comparison.

Configuration	Params (M)	Training (h)	FLOPs/Sample	Memory (GB)
Standard diffusion framework	64.47	8.5	3.2×10^8	4.8
ER-CMGI framework	16.63	1.3	7.1×10^7	1.2
Improvement	-74.2%	-84.7%	-77.8%	-75.0%

contributes to the model's efficient runtime behavior while preserving functional integrity.

To demonstrate the efficiency improvements achieved by our lightweight design, Table 9 presents a comprehensive comparison between our full ER-CMGI framework and standard diffusion-based approaches. The results highlight that our framework achieves significant efficiency gains through the integration of lightweight component design and optimized diffusion mechanisms. These improvements include reduced parameter overhead, simplified architectural complexity, and efficient cross-modal generation strategies.

The comparison demonstrates that our ER-CMGI framework achieves 74.2% parameter reduction (16.63M vs. 64.47M), 84.7% training time reduction (1.3 h vs. 8.5 h), 77.8% FLOPs reduction, and 75.0% memory reduction (1.2 GB vs. 4.8 GB) compared to standard diffusion-based frameworks while maintaining classification performance.

The comparative analysis demonstrates the superior efficiency of our ER-CMGI framework across multiple computational metrics. Compared to standard diffusion-based frameworks, ER-CMGI achieves a 74.2% reduction in model parameters (16.63M vs. 64.47M), an 84.7% decrease in training time (1.3 h vs. 8.5 h), along with 77.8% fewer FLOPs and 75.0% lower memory consumption (1.2 GB vs. 4.8 GB), all while maintaining competitive classification performance. These improvements translate into significant advantages for practical deployment scenarios. The lightweight design enables highly efficient training, reducing training time by 84.7%, which facilitates faster model iteration and deployment.

From an inference perspective, the architecture is optimized to maintain low computational overhead without sacrificing processing throughput. The asymmetric design ensures that the cross-modal generators – comprising only 20.3% of the total parameters – are primarily active during training. As a result, the deployed model utilizes only 13.24M active parameters and occupies a compact size of 33.3 MB under FP16 precision. Moreover, the model converges stably within 20 training epochs, further enhancing its suitability for resource-constrained environments.

5. Discussion

5.1. Limitations

While our experimental datasets provide valuable validation for the proposed approach, we acknowledge several inherent limitations that may affect real-world applicability. Dataset D1 offers comprehensive coverage of traditional protocols and malware families, while Dataset D2 incorporates recent TLS-encrypted traffic (2018–2021), enabling assessment across different temporal periods and encryption

paradigms. However, these datasets may not fully encompass the complexity of modern network environments, including emerging protocols (e.g., QUIC, HTTP/3), diverse geographical sources, and large-scale enterprise deployment scenarios. To address these limitations, we employ cross-dataset validation and comprehensive preprocessing normalization to reduce dataset-specific biases. Future work will investigate performance on larger-scale datasets and emerging protocol types to further validate the practical applicability of our approach.

The framework also exhibits sensitivity to multiple hyperparameters, including diffusion timesteps, entropy mixing coefficients, and loss weighting factors, requiring careful domain-specific optimization for different network environments and attack types. Furthermore, the current implementation is specifically designed for byte sequences and packet length sequences, and extension to additional modalities would require architectural modifications. A critical architectural limitation emerges from the byte encoder design, where the majority of model parameters are concentrated in a single large fully connected layer that processes flattened CNN feature maps. This design creates parameter inefficiency and potential information loss during aggressive dimensionality reduction, while constraining deployment on resource-limited environments.

In addition, while our approach incorporates basic anonymization mechanisms (removing IP/MAC addresses), it currently lacks formal privacy guarantees such as differential privacy or secure multi-party computation. The flow-level analysis approach, although more privacy-friendly than deep packet inspection, still processes raw network traffic that may contain implicit behavioral patterns. Future work could explore integrating differential privacy techniques by adding calibrated noise to latent representations or adopting federated learning approaches for distributed privacy-preserving intrusion detection, though such enhancements may require balancing privacy protection with detection accuracy.

Despite these limitations, the theoretical foundation and experimental results demonstrate the effectiveness of entropy-guided cross-modal generation for encrypted traffic analysis, providing a solid foundation for future improvements and practical deployment considerations.

5.2. Generalizability of the hybrid entropy strategy

The hybrid entropy strategy proposed in this work demonstrates potential for extension beyond network traffic analysis to other multimodal learning domains. The core principle – dynamically weighting modalities based on their information-theoretic properties while respecting modality-specific entropy characteristics – can be adapted to various scenarios where heterogeneous data sources exhibit different uncertainty patterns.

In audio-visual multimodal analysis, for instance, audio features often exhibit higher entropy due to temporal variability and noise, while visual features may show more stable patterns in certain contexts. The hybrid entropy approach could adaptively balance these modalities by applying different entropy weighting strategies: favoring high-entropy audio features during dynamic scenes (where audio provides discriminative information) while prioritizing low-entropy visual features during stable scenes (where visual consistency indicates reliable information). Similarly, in medical multimodal analysis combining imaging and sensor data, different modalities may exhibit varying reliability patterns across patients or conditions, making adaptive entropy-based fusion particularly valuable.

6. Conclusion

To address the challenges of dynamic feature fusion and cross-modal semantic alignment in encrypted traffic analysis, we propose the ER-CMGI. The framework integrates the probabilistic generation capabilities of diffusion models with information-theoretic optimization, enabling adaptive learning of multimodal representations. By introducing a lightweight hybrid architecture that combines diffusion models and variational autoencoders, our approach facilitates more coherent interaction between byte and length modalities, thereby overcoming the limitations of static fusion strategies and misaligned latent spaces commonly found in traditional methods.

We further propose an entropy guided fusion mechanism that dynamically balances the contribution of each modality based on its intrinsic information content. By adjusting the fusion weights based on the uncertainty and richness of individual modal representations, this method achieves better integration of supplementary information while maintaining model interpretability. The performance of adaptive weighting strategies has been demonstrated in complex classification scenarios, where modalities may exhibit varying degrees of discriminative ability. While the current framework shows promising results, future research directions should address computational efficiency through model compression and adaptive inference strategies, enhance robustness by developing adaptation mechanisms for emerging protocols such as QUIC and HTTP/3, and conduct large-scale validation studies in production environments to further demonstrate practical applicability in real-world deployment scenarios.

CRedit authorship contribution statement

Xiangbin Wang: Writing – original draft, Software, Methodology, Conceptualization. **Qingjun Yuan:** Writing – review & editing, Methodology, Data curation. **Wentao Yu:** Visualization, Validation. **Qianwei Meng:** Software, Formal analysis. **Siqi Lu:** Supervision, Funding acquisition. **Wenqi He:** Resources, Formal analysis. **Chunxiang Gu:** Writing – review & editing, Project administration. **Yongjuan Wang:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.inffus.2025.103581>.

Data availability

Data will be made available on request.

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Xiangbin Wang received the M.Eng. degrees from Information Engineering University. He is currently a graduate student studying for a Ph.D. in Information Engineering University, China. His research interests include network security and network traffic analysis.

Qingjun Yuan received the Ph.D. degree from Information Engineering University. He is currently a lecturer at Information Engineering University. His research interests include network security and side channel attack.

Wentao Yu received the Ph.D. degree from Information Engineering University. He is currently a lecturer at Information Engineering University. He is also a postdoctoral researcher at the Institute of Computing Technology, Chinese Academy of Sciences, and his research interests include artificial intelligence and knowledge engineering.

Qianwei Meng is currently pursuing the Ph.D. degree at Information Engineering University. His research interests include network security and unknown traffic detection.

Wenqi He received the M.S. degree in statistics from Zhengzhou University. Her research interest is network security.

Chunxiang Gu received the Ph.D. degree from Information Engineering University. He is currently a professor at Information Engineering University. His research interests include network security and cryptographic analysis.

Yongjuan Wang received the Ph.D. degree from Information Engineering University. She is currently a researcher at Information Engineering University. Her main research interests include cryptographic analysis and zero trust security.